Problem Set 5

Directions

Submit two files on Moodle before the deadline:
[1] your **solution set** as a *.pdf [2] the **R script** as a *.R type file

<u>For both, the first line should include your student ID and the assignment number</u>
If you worked with one or two other students write “group:” and then their student ID numbers
[1] the **solution set** should include for each answer:

- the question number/part
- your answer in words
- the command in R that was used (if relevant)
- the output from Rstudio (if relevant)

[2] the **R script** should include:

- all the commands you used to generate the solutions
- individually written comments that explain your code to the grader

Labor unions and product recalls

In a recent paper, Kini, et. al. (2023) study the effect of labor unions on product quality failures. They find unionization has a negative effect on the cumulative number of product recalls 3 years after the unionization. A number of federal agencies are able to require companies to issue product recalls due to concerns over consumer safety; they focus on food, drug, medical device, vehicles, and consumer products such as children’s products, household appliances, heating and cooling equipment, home furnishings, hardware and tools, and yard equipment among others.

For a firm’s workers to unionize, special secret ballot elections are held and over 50% of workers employed at the firm need to vote for unionization. Often multiple election rounds are held at the same firm in the same year. The data is *not* a panel dataset

You have access to data on the union elections conducted at 382 firms over 10 years from Kini, et. al. (2023). The dataset is called **union.data** in the *.Rdta data file on moodle.

- **firm_ID**: a unique identifier of the firm
- **year**: the year the election took place (2003 – 2013)
- **pct_vote_union**: the percent of workers who voted in favor of forming a union
- **union_won**: indicates the vote percent was greater than 50% and the union won.
- **RTW_state**: indicator that the firm is located in a “right to work” state, meaning that workers can opt to work at the firm without joining the union
- **industry_ID**: ID indicating the industry the firm is in
- **has_recall_1yr**: indicates the firm had at least one product recall in the last year
- **has_recall_2yr**: indicates the firm had at least one product recall in the last 2 years
- **has_recall_3yr**: indicates the firm had at least one product recall in the last 3 years

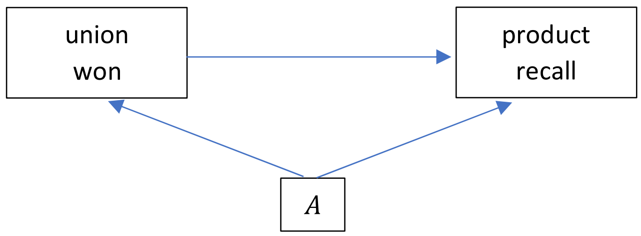
<u>Research question</u>: **What is the effect of unionization on product recalls?**

1. **Baseline model**. Consider the following baseline model (B):

$$has.recall_i = \beta_0 + \beta_1 union.won_i + u_i$$

Estimate the model three different times, each time using a different outcome variable has_recall_3yr, has_recall_2yr, or, has_recall_1yr. Report your results. How do your results change for each of the outcome variables. Provide an explanation for why this might be the case.

2. **Causality**. Consider the diagram below. What is something unobserved which might fall into category A? How does this affect our interpretation of β_1 in the baseline model?



3. RDD setup. The researchers employ a regression discontinuity design to study the effect of unionization on whether a firm has a product recall in the last 3 years.
The outcome variable is has_recall_3yr.

a. What is the "running variable" for this study? What is the "treatment" variable? What is the cutoff that determines "treatment"?

b. Plot the running variable on the x-axis, and the treatment variable on the y-axis. Add a vertical line where the cutoff falls. What type of RDD study is this?

4. RDD. Consider the RDD model below (RDD1).

$$\begin{aligned} has.recall.3yr_i = \beta_0 + \beta_1 (running - cutoff)_i + \beta_2 treatment_i \\ + \beta_3 (running - cutoff)_i \cdot treatment_i + u \end{aligned}$$

a. Which coefficient measures the treatment effect we are interested in? What does it represent?

b. Estimate the model (use the variables you answered from **3**). Use the full sample of data. Report your results.

c. Estimate the same model from 3b. Use only a subset of observations: those with running variable that is within 0.1 of the cutoff. Report your results.

d. Which of your results (b or c) is the one you should use and why?

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5. RDD with polynomial. Consider the RDD model below (RDD2):

$$has.recall.3yr_i = \beta_0 + \beta_1(running - cutoff)_i + \beta_2(running - cutoff)_i^2$$
$$+\beta_3 treatment_i + \beta_4(running - cutoff)_i \cdot treatment_i$$
$$+\beta_5(running - cutoff)_i^2 \cdot treatment_i + u_i$$

a. Estimate the model RDD2. Based on your answer for **4d**, estimate the model once either for the full dataset or using a subset of observations (within 0.1 of the cutoff). Report your results

b. How does your answer change (if at all) between RDD1 and RDD2? Why might this be the case?